

Beyond a Single Backbone: A Preregistered Multi-Backbone Test of a Geometry-Mediated Account of Distance-Based Reliability Estimation in Dermatology

[Author list]*

Abstract

Distance-based reliability estimators flag inputs that a deployed classifier should not be trusted on, and their behaviour under acquisition shift has been linked to the geometry of the frozen representation they operate on. A preregistered causal study established that link on a single dermatology foundation model, where condition number was the only geometry metric significantly associated with estimator performance. Whether that account describes representations in general or that representation in particular cannot be determined from one backbone: with a single instance, the two produce identical evidence.

We applied the same intervention ladder—dose ladder, matched control arms, and implementation and manipulation gates—independently to ten frozen backbones, sampled as five representation families with two instances each so that within-family variance could be measured rather than assumed. The analysis was preregistered and frozen in a single commit before any cross-backbone association was computed. Primary inference resampled whole random seeds rather than individual observations, since observations sharing a seed share an adapter and a training trajectory.

We found no confirmatory evidence that the association generalizes: no backbone cleared the Holm-corrected threshold, and the pattern across families was heterogeneous. Six of ten backbones nonetheless showed $|\tau| \geq 0.29$, in both directions, so this is not evidence of absence. On the original backbone the original result reproduced exactly ($\tau = +0.5576$, difference 0.0000), which removes the analysis pipeline as an explanation for the null. Evaluated under seed-level permutation, that same τ gives $p = 0.13$; the two inference procedures, differing only in their unit of resampling, disagreed on six of ten backbones.

With five seeds, the permutation distribution and the Holm correction together severely limit attainable family-wise significance, and the null should be read with that constraint. We report a boundary of validity rather than a refutation, and note that the earlier inference is not robust to changing the unit of resampling.

Keywords: reliability estimation; distribution shift; representation geometry; preregistration; replication; dermoscopy

1 Introduction

A skin-lesion classifier deployed outside the setting it was trained in will meet images it should not be trusted on: a different camera, a different clinic, a different population. The practical safeguard is not a better classifier but a *reliability estimator*—a mechanism that flags such inputs before a prediction is acted upon. Distance-based estimators are the usual choice, because they operate on a frozen representation, require no retraining, and depend on no classifier head. Mahalanobis distance to class-conditional centroids is the canonical instance.

*Correspondence: [email]

Whether these estimators actually survive acquisition shift has proved harder to settle than early results suggested, and the difficulty is instructive. On dermoscopy-to-smartphone shift, Mahalanobis distance separated in-distribution from shifted inputs at AUROC 0.97 while softmax confidence collapsed [1]. That robustness did not survive scrutiny: after domain-adversarial shortcut removal, the same estimator on the same shift fell to approximately 0.40 [2]—worse than chance, and therefore not merely a loss of discriminative power. The natural reading is that the original robustness had been reading a shortcut rather than the shift itself.

The reading that followed was less natural and more interesting. If the estimator fails because the domain information has been removed, a domain probe applied to the same embeddings should also fail. It does not. A linear probe still decoded domain at 0.72–0.81 AUROC from embeddings on which eight distance and density estimators performed at or below chance [3]. The information is present. The estimators cannot reach it.

That is a statement about the *geometry* of the representation rather than its information content, and it converts a negative result into a mechanistic question: which property of an embedding space determines whether a distance-based estimator can exploit the structure it contains?

A preregistered causal study addressed that question on a single frozen dermatology foundation model [4]. Rather than comparing representations observationally, it intervened: a small adapter on the frozen output embedding, trained to suppress a closed-form nuisance direction, with a dose ladder obtained by post-hoc interpolation and matched control arms that isolate capacity from the orthogonality objective. Among the preregistered geometry metrics, condition number was the only one Holm-significantly associated with estimator performance; local intrinsic dimensionality and within-class spectral decay were not.

That study was explicit about what it could not establish. Its own analysis flagged that treating “architecture” as a causal category on the basis of a single instance is unfalsifiable by design: with one backbone, a property of *representations* and a property of *that representation* produce identical evidence, and within-category variance is never measured. It deferred the multi-backbone test to future work for resource reasons rather than logical ones.

Filling that gap is not simply a matter of running more models. Large-scale multi-backbone studies of representation quality exist, but they are correlational: they compare backbones as found, so any association between geometry and estimator behaviour is confounded by everything else that differs between them. What the question requires is the *same* intervention applied *independently* within each backbone, so that the causal claim is tested once per representation and then compared across representations—with backbone identity modelled as a source of variance rather than assumed away.

This study does that. We apply the full intervention ladder—dose ladder, control arms, implementation and manipulation gates—independently to ten backbones, sampled as five representation families with two instances each so that within-family variance is measurable rather than assumed. The families span supervised CNNs, medical and general self-supervised encoders, and medical and general vision–language models. A segmentation encoder was included separately as a declared architecture-portability probe.

The analysis was preregistered and frozen before any cross-backbone association was computed. The population, the covariate definition, the confirmatory family, the inference procedure, the outcome taxonomy and the stopping rule were each fixed in advance, and the freeze is a single commit whose timestamp precedes every result reported here. Two features of that protocol matter for reading what follows: the primary inference resamples whole random seeds rather than individual observations, because observations sharing a seed share an adapter and a training trajectory; and a backbone whose manipulation check fails is recorded as *not testable* rather than as evidence against the hypothesis.

We report three things. Under the preregistered protocol we found no confirmatory evidence that the geometric relationship generalizes across backbone families. On the original backbone,

Table 1: Sampled backbones. Dimensions are pre-projection.

Family	Instance 1	dim	Instance 2	dim
CNN, supervised	ResNet-50	2048	EfficientNet-B3	1536
Medical, image-only SSL	PanDerm	1024	UNI	1024
Medical, vision-language	BiomedCLIP	768	MONET	1024
General, image-only SSL	DINOv3	1024	MoCo v3	768
General, vision-language	OpenCLIP	768	SigLIP	1024

the original result reproduced exactly—to four decimal places—which removes the pipeline as an explanation for the first finding. And the two inference procedures, differing only in their unit of resampling, disagreed on six of ten backbones, including on the exact effect size the original study reported.

2 Methods

This section is derived from the frozen protocol and preregistration and introduces nothing absent from them.

2.1 Data

ISIC 2019 ($n = 25\,331$, in-distribution) and PAD-UFES-20 ($n = 2298$, acquisition-shifted). Split seed 42, identical to the preceding papers in the programme and verified as such rather than assumed: training population $n = 16\,211$, in-distribution evaluation $n = 5067$, shifted evaluation $n = 2298$, evaluation pool $n = 7365$.

Split assignments are pinned by a digest over image identifier, label, domain and partition, sorted by identifier—deliberately excluding file paths, which are machine-local and not split variables. The digest is verified identical when computed from the working metadata and from the frozen reference embeddings of the previous study.

2.2 Backbones

Ten backbones, sampled as five representation families with two instances each, chosen to measure within-family variance rather than to represent architectures individually (Table 1). A segmentation encoder (MedSAM) was included as a declared architecture-portability probe, outside the family contrast and excluded from every denominator.

All embeddings are taken pre-projection. Every checkpoint is pinned to an immutable revision; library versions are pinned and recorded per run.

2.3 Intervention

Backbones remain frozen. A bottleneck adapter acts on the output embedding,

$$z' = z + W_2 \sigma(W_1 z), \quad (1)$$

with W_2 zero-initialised so the adapter is the identity at initialisation and σ matched to each backbone’s own activation. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{proj}} \cos^2(z', w), \quad w = \frac{\mu_{\text{ISIC}} - \mu_{\text{PAD}}}{\|\mu_{\text{ISIC}} - \mu_{\text{PAD}}\|}, \quad (2)$$

where the nuisance direction w is computed in closed form per backbone and is never learned. The dose ladder $z'(\alpha) = z + \alpha \Delta z$ for $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is post-hoc interpolation from a single trained adapter, not five separate training runs.

The bottleneck width and λ_{proj} are selected per backbone from a pre-committed grid—the smallest value passing both gates, using gate outcomes only and never outcome results.

2.4 Gates

Gate 0 (implementation integrity) and Gate 1 (manipulation check) are eligibility criteria, scored before and independently of the association. A Gate 1 failure records a backbone as *not testable*; it is never counted as falsification.

2.5 Analysis population

Per backbone, $n = 30$: the intervention arm at $\alpha \in \{0.25, 0.5, 0.75, 1.0\}$ across five seeds, plus the adaptation arm’s linear-probe and partial-fine-tuning rungs across five seeds each. The full-adapter rung is excluded, as it reuses the control arm’s checkpoints.

The dose ladder alone is not sufficient: on the intervention arm only ($n = 20$) the same association gives $\tau = 0.242$, $p = 0.146$.

2.6 Confirmatory analysis

Covariate. $\kappa_{\text{primary}} = \lambda_1/\lambda_k$ over the descending eigenvalues of the pooled within-class covariance Σ_W , unregularised, with $k = 256$ fixed across backbones. The prior study’s definition κ_{prior} —full-dimension, regularised, from the precision matrix—is reported alongside for direct replication comparison only.

Because κ is scale-invariant, $\kappa(c\Sigma_W) = \kappa(\Sigma_W)$, no normalisation is applied; the confound is the *absolute* regulariser, which does not rescale with the matrix, and κ_{primary} takes no inverse and therefore needs none. Measured across the ten backbones, κ does not grow with embedding dimension (Spearman -0.308 , $p = 0.386$); what κ_{primary} demonstrably does is make κ commensurable, compressing the cross-backbone spread from $3918\times$ to $5.2\times$.

Outcome and test. Mahalanobis AUROC; Kendall’s τ per backbone. Primary inference is a seed-level permutation test, permuting whole seeds; the parametric τ is secondary, and where the two disagree the permutation result governs. This was fixed before either was computed.

Multiple testing. The confirmatory family is $\kappa_{\text{primary}} \times \text{Mahalanobis} \times 10 \text{ backbones} = 10$ tests, Holm–Bonferroni, $\alpha = 0.05$ family-wise. Cosine-to-centroid, k -nearest neighbours and kernel density estimation are secondary and lie outside the correction.

Classification. Two tiers. Tier 1 is defined on the count of Holm-significant same-direction backbones alone; Tier 2 describes the family pattern. Results are reported as a pair.

3 Results

Preregistered content only. No interpretation appears in this section.

3.1 Testability

All ten family members passed both gates: $T = 10$, all five family cells intact. The preregistered testability condition ($T \geq 5$ and at least two complete cells) was met, and all ten joined the analysis population at 30/30.

The declared probe passed Gate 0 on all twelve grid configurations and failed Gate 1 on all twelve. It is recorded *not testable* and excluded from every denominator.

Table 2: Confirmatory analysis. τ is identical under both procedures; only the p -values differ. AUROC range is reported beside every τ so a rank association is not read as a difference in estimator performance.

Backbone	τ	p_{perm}	p_{param}	p_{Holm}	Sig.	AUROC range
ResNet-50	+0.0306	0.8500	8.16×10^{-1}	1.0000	no	0.7786–0.7800
EfficientNet-B3	−0.5153	0.3333	8.57×10^{-5}	1.0000	no	0.8181–0.8213
PanDerm	+0.6047	0.0583	4.04×10^{-6}	0.5833	no	0.9983–0.9988
UNI	−0.2988	0.8250	2.27×10^{-2}	1.0000	no	0.9722–0.9734
BiomedCLIP	−0.3271	0.4250	1.27×10^{-2}	1.0000	no	0.9680–0.9766
MONET	−0.1859	0.1417	1.57×10^{-1}	1.0000	no	0.9926–0.9934
DINOv3	+0.3082	0.3083	1.88×10^{-2}	1.0000	no	0.9819–0.9851
MoCo v3	−0.3788	0.2167	3.88×10^{-3}	1.0000	no	0.9749–0.9752
OpenCLIP	+0.0259	0.9083	8.44×10^{-1}	1.0000	no	0.9781–0.9791
SigLIP	−0.0447	0.8083	7.33×10^{-1}	1.0000	no	0.9655–0.9668

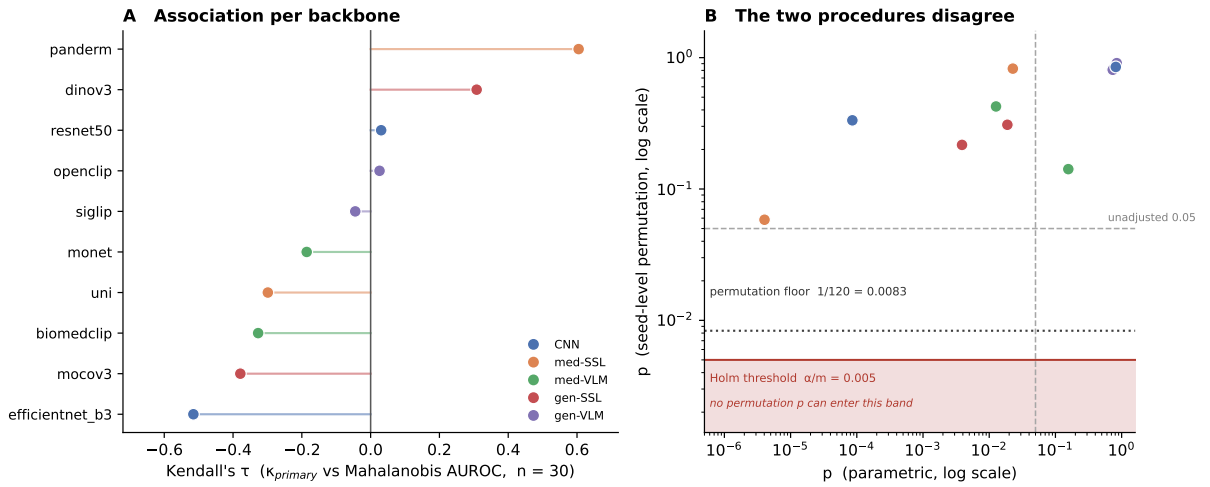


Figure 1: (A) Association per backbone, ordered by τ and coloured by representation family. (B) The two inference procedures plotted against each other on logarithmic axes. The permutation floor $1/120$ and Holm’s first threshold $\alpha/m = 0.005$ are marked; the floor lies above the threshold, and the shaded band is unreachable by any permutation p -value at five seeds.

3.2 Per-backbone association

Table 2 reports κ_{primary} against Mahalanobis AUROC, $n = 30$ per backbone, with Holm–Bonferroni applied to the permutation p -values and the AUROC range beside every τ . Figure 1A shows the same values ordered by τ .

3.3 Outcome

$S = 0$; Tier 1 is D and Tier 2 is heterogeneous. The reported pair is **Outcome D, heterogeneous**. All five family cells were 0/2; mean permutation τ was negative in four families, with medical image-only SSL at +0.1529.

3.4 The two inference procedures disagreed on six of ten backbones

With identical τ and unadjusted $\alpha = 0.05$, the permutation and parametric tests disagreed for EfficientNet-B3, PanDerm, UNI, BiomedCLIP, DINOv3 and MoCo v3 (Table 2; Figure 1B). Per the preregistration, the permutation result governs.

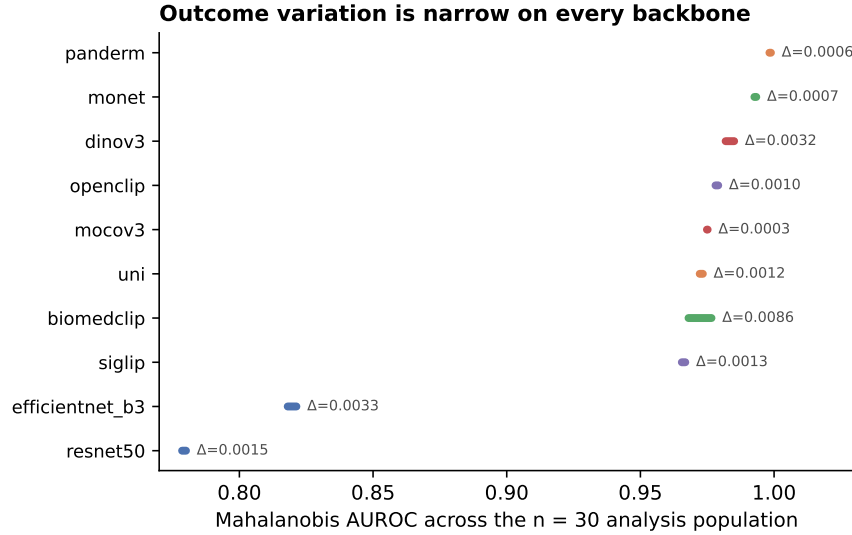


Figure 2: Mahalanobis AUROC across the $n = 30$ analysis population, per backbone.

3.5 Replication of the original result

On the original backbone, using the original covariate definition and the same analysis population, $\tau = +0.5576$ against a published $\tau = +0.5576$, a difference of 0.0000. Under the seed-level permutation test the same τ gives $p = 0.1333$.

3.6 Sensitivity

The preregistered sensitivity analyses agree with the primary result. κ_{primary} at $k \in \{128, 256, 512\}$ gives $S = 0$ and Tier 1 D at every k ; κ_{prior} likewise.

3.7 Outcome variation

Figure 2 shows the Mahalanobis AUROC range per backbone. The widest is 0.0086 and the narrowest 0.0003. Kendall’s τ is scale-free and is unaffected by this; the ranges are reported so that a rank association is not read as a difference in estimator performance.

3.8 Secondary scorers

Cosine-to-centroid, k -nearest neighbours and kernel density estimation were computed in full and lie outside the confirmatory correction, as preregistered. They are reported in Supplementary Table S1 and support no claim in this section.

4 Discussion

4.1 The preregistered result

Under a preregistered evaluation protocol, we found no confirmatory evidence that the geometric relationship reported in the previous study generalizes across backbone families. No backbone cleared the Holm-corrected threshold, the outcome was classified as D, and the family pattern was heterogeneous.

Two scope conditions belong in the same breath as that claim, because without them it reads as something stronger than the data supports.

This is not evidence of absence. Six of ten backbones showed $|\tau| \geq 0.29$. What distinguishes them is not that the associations were small but that they pointed in both directions: four positive, six negative, with the largest of each sign at $+0.60$ and -0.52 . A body of near-zero associations would have been evidence against the hypothesis. A body of substantial associations without a consistent sign is not the same thing, and should not be reported as though it were.

The procedure’s resolution was limited. With five seeds the seed-level permutation distribution admits 120 arrangements, so its p -values are quantised at multiples of $1/120 \approx 0.0083$, while Holm’s first threshold across ten tests is 0.005. The combination of five clusters and Holm correction severely limits attainable family-wise significance. This is a property of the design, fixed before any result was seen, and it constrains what a null can be taken to mean here. A reader who takes Outcome D as a measurement of the world rather than partly a measurement of the procedure will overread it.

4.2 Replication

On its original backbone, using the original covariate definition and the same analysis population, the previous result reproduced exactly: $\tau = +0.5576$ against a published $\tau = +0.5576$, a difference of 0.0000.

This forecloses the explanation a null result usually invites. The same code that returned no confirmatory evidence across ten backbones reproduced the original finding to four decimal places on the one backbone where a published value exists to check against. Whatever accounts for the first result, it is not that the pipeline was wrong.

The same τ , evaluated under the seed-level permutation test, gives $p = 0.1333$. Nothing about the data or the effect size changed; the unit of resampling did. Observations sharing a random seed share an adapter, a training trajectory and an initialisation, and treating the thirty rows as thirty independent draws is what separates the two p -values.

We therefore report that the earlier inference is *not robust to changing the unit of resampling*. That is a statement about a procedure, and it is checkable. It is not a claim that the earlier finding was wrong: a permutation test over five clusters has little power, and $p = 0.1333$ records a failure to resolve rather than a demonstration of absence. Both readings—that the effect is real, and that the earlier significance was overstated—remain open on this evidence, and we do not choose between them.

4.3 Boundary of validity

The two studies answer different questions. The first asked whether the phenomenon exists on one backbone. This one asks whether it holds across many. A negative answer to the second does not retract the first, and the exact replication above is what makes that separation credible rather than merely rhetorical. What this study establishes is a boundary, not a refutation.

The boundary extends to the protocol as well as to the finding. The manipulation check inherited from the earlier study accepts a geometry change only in the decreasing direction. Across the present backbone set, within-class spectral decay moved *upward* under intervention in eight of ten cases. With a single backbone, “the intervention decreases this quantity” and “the intervention changes this quantity” were indistinguishable; with ten they are not. The declared portability probe—a segmentation encoder, deliberately the most structurally distant backbone in the set—passed the implementation gate on all twelve grid configurations and failed the manipulation gate on all twelve, and is recorded *not testable* rather than as a negative result. That is the outcome the probe was included to make possible.

Exploratory analyses suggested that variability attributable to seed and variability attributable to the intervention were distributed differently across quantities, motivating future work on experimental design.

4.4 Limitations

Five seeds. The resolution limit above follows directly from the number of seeds, not from the choice of a permutation test. A cluster-level procedure with more clusters would not have this constraint.

Two instances per family. Family-level statements rest on two models each, which is the minimum that permits within-family variance to be estimated at all and is far from the minimum that would make it precise.

A one-sided manipulation criterion. The inherited gate accepts only decreasing geometry change, while the protocol text describes dose-dependence without specifying direction. The two do not agree. We report the discrepancy rather than repairing it: by the time it was identified it was already known which backbone the repair would change, so amending the criterion would have been outcome-contingent.

Non-uniform artifact release. Several checkpoints carry non-commercial licences and one prohibits redistribution of derivatives, so released artifacts are represented by checksums where the licence requires it. Reproduction from a clean clone is therefore not uniform across backbones.

Saturated outcome. Mahalanobis AUROC varied within a narrow band on every backbone. Kendall’s τ is scale-free and is unaffected by this, but a rank association over differences of this size should not be read as a practically meaningful difference in estimator performance. Every τ in Section 3 is reported with its AUROC range for that reason.

4.5 Future work

The design constraint identified above is specific and actionable: at a fixed budget, the binding resource for this class of study is the number of independent seeds rather than the number of rungs on the dose ladder. We do not develop that further here.

Data and code availability

Both datasets are public. The split is pinned by a digest over assignment columns rather than over the metadata file, so it is verifiable on any machine.

The repository contains the protocol, the preregistration frozen as a single commit, the analysis code, the per-backbone result tables, and a run manifest recording the commit, configuration hash, checkpoint revision and library versions for every run. Every checkpoint is pinned to an immutable revision rather than to a branch. A single script regenerates every number in Section 3 from version-controlled inputs.

A decision log covering the full design and analysis history accompanies the repository. It records each decision with the date, the commit, and whether outcome data had been seen at the time—the last of which is what distinguishes a preregistered choice from a post-hoc one, and is the reason the exploratory analyses that were run and deliberately excluded are recorded rather than omitted.

Release is stratified by checkpoint licence. Three of the ten backbones carry non-commercial terms, one prohibits redistribution of derivatives, and one is withheld pending a legal question about ShareAlike obligations. Artifacts derived from those backbones are represented by checksums. Reproduction from a clean clone is therefore not uniform across backbones, and we state this rather than implying a uniform reproduction path that does not exist.

Embeddings and adapter checkpoints total tens of gigabytes and are not held in version control; their checksums are.

Deviations from the preregistration

None.

References

- [1] TODO. Mahalanobis distance is robust to acquisition shift where softmax confidence collapses in dermoscopic lesion classification. *TODO*, TODO. Paper 1 of the programme.
- [2] TODO. Csg-skin: shortcut removal collapses distance-based reliability estimation under acquisition shift. *TODO*, TODO. Paper 2 of the programme.
- [3] TODO. Dst-skin: domain information survives the collapse of distance and density estimators. *TODO*, TODO. Paper 3 of the programme.
- [4] TODO. A preregistered causal test of a geometry-mediated account of distance-based reliability estimation. *TODO*, TODO. Paper 4 of the programme; the study replicated here.